Chapter 21

Application of nanosensors in the food industry

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21.1 Introduction

Nanotechnology and nanoscience have been instrumental in several fields such as energy generation, medicine, computer sciences, communication, and the food industry. With the advancement in the food industry and in the agricultural sector the diversity and amount of food have dramatically enlarged. This inevitably brings the safety of our food on top priority as it directly influences the socioeconomic factors and attracts global attention. Nanotechnology has a wide range of applications in the food industry. Nanostructure constituents vary differently in their structural elements, clusters, amorphous and crystallites in the form of nanoparticles (NPs), nanoclusters, nanothin films, nanorods, nanotubes, and magnetic beads in the range of 1−100 nm as shown in Fig. 21.1 [1].

The molecular chemical components related to food safety are varied and complex [2,3]. The protein and the DNA modifications within the biological environment have made the detection of pesticides, veterinary drug residues, use of rampant additives, heavy metals, organic compounds, pathogens, and toxins extremely difficult. These modified species ultimately give rise to foodborne diseases that severely affect the health as well as the overall growth of food business concerns [4]. The outstanding physiochemical properties and the antimicrobial benefits of nanosized chemical species make them the most suitable candidates for the treatment of pathogenic conditions in health care, crop protection, water treatment, food safety, and food preservation [5]. Nanotechnology has played an instrumental role in the research and development for the large-scale production of agricultural products, processed edibles, as well as for packaging procedures across the world. Several findings affirm that these nanomaterials successfully modify safety methods for edibles by uplifting the effectuality of packaging procedures, improve shelf life, as well as the nutritional content of the food products without changing their desired taste and physical characteristics [6]. Nanotechnology affords as an important technique for the biosensing potential for food applications. The successful demonstration of PDMS/paper/glass hybrid microfluidic biochips performing the one-step “turn-on” homogeneous assay for multiplexed pathogen detection using functionalized graphene oxides (rGO) prove the enhanced sensitivity of these nanomaterials.

Although, a large number of bioassay tools have flooded the market, these state-of-the-art systems were able to satisfy many practical issues and to address the ever-growing efforts in order to develop new biodetection approaches with ultimate ultrasensitivity helping to improve the limit of detection to very few molecules. It includes all food-related activities such as preparation, handling, storage, and food distribution, as well as simultaneously includes the detection, control, and prevention of any kind of intentional or unintentional spoilage. Thus, to assure food safety, food industries must hold up a highly impressive sanitizing mechanism to reduce the population of pathogens and spoilage

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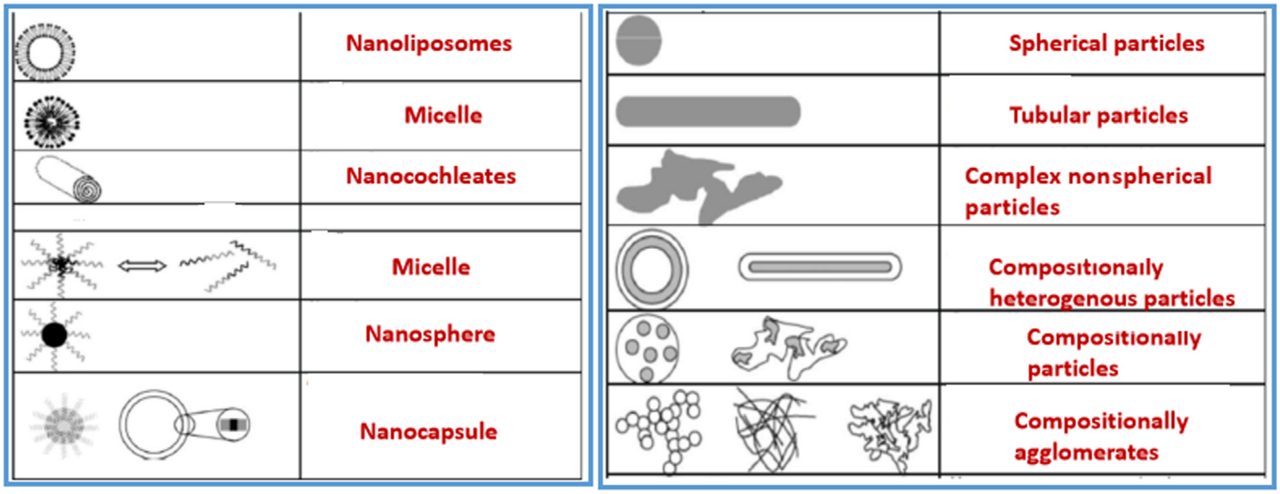

FIGURE 21.1 Different shapes of nanoparticles. *Reprinted with permission from H. Bouwmeester, et al., Review of health safety aspects of nanotechnologies in food production, Regul. Toxicol. Pharm. 53 (1) (2009) 52−62. © 2009 Elsevier.*

microorganisms on contact surfaces of food materials and within the food during its preparation to packaging stages. Various physical methodologies have been applied for reducing the microbial population in the food sector such as high-pressure spraying, pressurized steam, automated brushing, high-temperature sterilization, vacuum desiccation, and photolytic irradiation. Apart from physical and biological methods, chemical procedures have also been employed for sensing harmful pathogens through the biocidal application of phenolics, halogens such as iodine, biguanides, heavy metal contents, acid-anionic based quaternary ammonium constituents, sanitizers, aldehyde residues, and vaporized chemosterilants [7].

Global health concerns such as food poisoning is an important aspect in the developed as well as developing nations. Effective management and measures taken for early detection of food-poisoning conditions prevent any untoward outbreaks. Conventional laboratory methods applied for sensing of food-poisoning agents rely on culture growth, recognition of microbial causes [8], and detection of nucleic acid by polymerase chain reaction methods [9], immunological assays such as lateral flow immunoassay and enzyme-linked immunosorbent assay (ELISA) [8−13], and of toxins and chemical contaminants [10]. These conventional methodologies are specifically sensitive for the sensing of microbes and other known species. However, some of these procedures are difficult, inefficient, and require skilled laboratory professionals [11]. These challenges could be overcome by the union of conventional methods with nanotechnology. The successful application of their amalgamation can be seen in the 8-compartment systems for the fast detection of *Salmonella* bacterial colonies in food samplings. This procedure for bacteria detection was carried out on the sample loading through the use of magnetic beads within the loop-mediated isothermal amplification [12]. Biogenic amines are small aliphatic, aromatic, and heterocyclic molecules exhibiting biological assay. These are present in a wide array of food products of biological origin as well as in agitated food products. Biogenic amines are synthesized mainly by the unicellular microbes containing amino acid mainly histidine. In the view of their importance for human health and food safety, it is significant to monitor biogenic amine levels in foodstuff and drinks. Biogenic amines are the main cause of poisoning of food. The most frequent histamine poisoning is known as "scombroid fish poisoning," linked to eating of fishes intoxicated with high levels of histamine.

Through the previous decades, numerous positive steps have been taken toward the improvement of nanotechnology. The unparalleled properties of nanomaterials in conjugation with proper ligands provide unparalleled opportunities to create ultrasensitive detection methodologies. The increasing demand for strict testing and controlling harmful substances in food products directly affect the boom in the research of food-safety sensors [13]. Technological advances brought up by the conjunction of NPs and electrochemical optical detection modes offer fast, economical, and ultrasensitive methodologies for point-of-care testing. The key significant developments of improved electrochemical nanosensors for molecules, enzyme-based biosensors, gene-based sensors, and immunosensors were studied to diversify the interest of the readers with different concerns as shown in Fig. 21.2.

Many functional nanoproducts generate an integrated effect on catalytic and signal activities toward the specific molecule as well as advanced selectivity supported on specific identification among molecules. Thus a broad range encompassing the nanomaterial-based modified research collectively puts the electrochemical biosensors on the forefront in food safety [15].

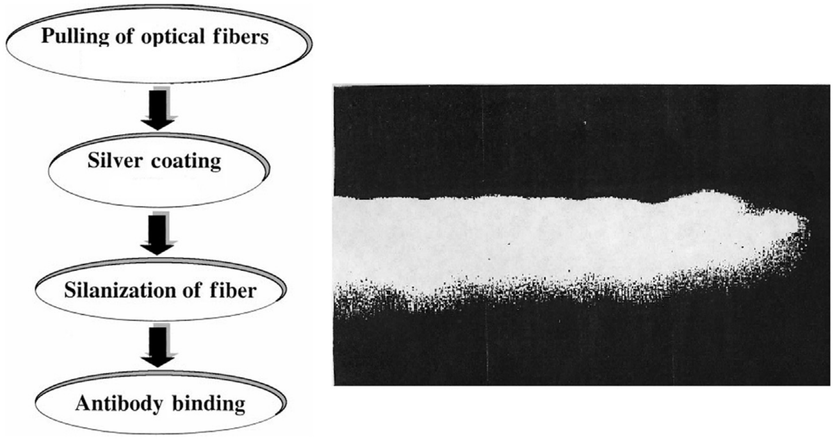

FIGURE 21.2 Schematic diagram for nanosensor fabrication along with SEM image of metal-coated nanofiber. *SEM*, scanning electron microscope. *Reprinted with permission from T. Vo-Dinh, B.M. Cullum, D.L. Stokes, Nanosensors and biochips: frontiers in biomolecular diagnostics, Sens. Actuators B: Chem. 74 (1–3) (2001) 2–11 [14]. © 2001 Elsevier.*

Another advancement in the application of nanomaterials within the food-safety methodologies is in the form of nanocantilevers. They are nanosensors composed of silicon-based materials exhibiting excellent microbial-sensing abilities. The enhanced biological-binding interactions among antigen with antibody or enzyme in association with substrate with effective physical and/or electromechanical methods enable these nanomaterials for excellent microbial detection. These nanosensors are extremely crucial in the understanding of interactions within molecules along with the detection of toxic chemicals and residual antibiotics in food items. Materials manufactured based on nanoelectromechanical systems (NEMS) are widely used for quality-control measures in food safety. They consist of modified transducers which effectively transform physicochemical stimuli into electrical pulses that serve as an important implement in preservation of food. One of the successful applications of this system is the use of digital transform spectrometer (DTS) incorporated with the enhanced technology of microelectromechanical systems produced by Polychromix, to detect the presence of *trans*-fat content in foods [16]. Such advantages of nanomaterials in food safety are already exploited in well-known organizations, such as LamdaGen and Nanolane, to ascertain the safety of their food products.

The value of nanotechnology in processes within the food industry can be judged through the consideration of its role in the betterment of the edible items in terms of (1) shelf life, (2) taste, (3) food presentation, (4) nutritional content of the edibles, and (5) texture. Apart from all the above points, it is commendable that nanotechnology has brought about significant alterations in food products by providing them with novel qualities. The microbiological aspect and nanotechnology-based methods have been useful in implementing proper food preservation, processing, quality monitoring, packaging, and storage, resulting in the overall enhancement of food quality and safety [17]. These applications, however, are not constricted to the utility of nanopackaging materials but expand the product-storage steadiness and use of nanocomponents with antimicrobial possess ions which finally result in the complete inhibition of the growth and detection of activities of spoilage by pathogenic microbes. Some important applications of nanotechnology for the benefits regarding the food safety comprise security of food from foreign pathogenic materials. These applications would be instrumental especially toward current and emerging technologies for food formulations, processing, and storage [18]. Packaging procedures hinder the consumers' sensory cognition, and thus they are advised to depend on sell-by dates, supported on platform of perfect assumptions, among the methods in which food would be contained or relocated. Nanosensors provide an alternative to this problem by altering the quality of the food items. Highly sensitive nanosensors have been fashioned, which can be incorporated in the packaging substances that would function as "noses" to discover the toxins discharged during food degradation [19]. Microfluidic devices are another alternative with highly sensitivity nanosensors, to rapidly detect pathogens. Advantages of these sensors are their miniature dimensions to detect related compounds quickly in tiny volumes justifying well spread use in various fields of medical, chemical, and biological analysis [14]. Many lawfully allowed additives in edible items are nitrites, acetic acid, citric acid, sulfites, sorbic acid, benzoic acid, sodium benzoate, and sodium propionate [20,21]. Identification of chemicals in food and methods to get rid of contaminants using nanotechnology will be discussed later.

21.2 Nanosensors for food safety

Recent developments in nanotechnology have completely revolutionized a large segment of the food sector and other industrial areas. This technology is hovering on the reduced-sized materials involving the constituent particles with the size in the range of 1−100 nm giving rise to a new class of materials with unique size-dependent properties. The increasing concerns of the consumer about food safety and its health impacts are compelling the researchers to develop new methods for a better quality of food without changing its nutritional value. The essential elements in the NP-based materials make nanotechnology the effective candidate in the food safety−related applications. The use of nanomaterials is on the rise in the food sector especially in the domains of food processing, functional food development, detection of foodborne pathogens, food microbiology, food packaging, and shelf life extension of edible products. Nanostructured ingredients and food nanosensing are the two main groups of nanotechnologies applied in the food sector. The entire spectrum of food processing to food packaging is enclosed within nanostructured ingredients. The properties of these nanostructures make them efficient to provide nutrients, permanency of the product packaging and mechanical strength, as well as they can serve as food additives, antimicrobial agents, etc. [22].

Extremely advanced sensing ability and other favorable properties of nanomaterials make them perfect representatives of biosensors. Quantification of available food constituents, identification of pathogens in food products and agriculture, and alerting the distributors and consumers on the safety aspect of food products are carried out using nanosensors or nanobiosensors [23]. The minute changes in the external environment such as microbial contamination, humidity fluctuation, and temperature in storage rooms are sensed by the nanosensors that are further indicated by their quick response mechanisms [1,24]. Various nanostructures, such as NPs, nanofibers, and nanorods, were examined for their potential in biosensing properties [25]. Out of all these, it was found that the thin film−based optical immunosensors were excellent in detecting the minute traces of microbial and cellular populations. On the surface of thin nanofilms or sensor chips the immobilization of certain protein content or antibodies is carried out, which gives rise to sensitive and fast-detection mechanism [26]. Effective sensing of foodborne pathogens, such as *Escherichia coli* and *Staphylococcus aureus*, has been carried out effectively by dimethylsiloxane microfluidic immunosensor with certain antibodies loaded on nanoporous alumina membrane [27]. Carbon nanotubes (CNTs), used to develop biosensors, are known for their facile- and low-cost capability toward sensor applications. The degradation of food products and the presence of microbial flora and fauna have effectively been sensed by these nanotube-based biosensors [28]. The change in the electric conductivity occurs when toxic waterborne antibodies get attached to these nanotubes and give the signal indication of waterborne pathogen presence [29]. The other few successful nanobiosensors are the use of quartz crystal microbalance-based electronic nose or tongue which detects and signals the presence of pungent gases released from the edible products [30,31]. Quartz crystal surfaces have been successfully used to detect various functional groups or biological residues such as enzymes, lipid molecules, amines, and diverse polymers [32]. Thus bringing biology and nanoscale technology on the same platform in the construction of sensors would sure enough reduce the response time to sense any potential hazard [33,34].

21.2.1 Detection of chemicals and contaminants

There are several conditions that cause the contamination of food, such as naturally produced toxicants due to bacteria and excessive use of chemicals, veterinary drugs, and pesticides that are added during food production. Detection of chemicals and contaminants in food products is the prerequisite to prevent harmful effects on public health. For this reason, analysis of process and quality control of food products are essential in food industries. Sensing of chemical and biological contaminants in food is crucial as they cannot be seen directly like physical contaminants. Along with sensing of sugar, alcohols, amino acids, flavors, and sweeteners, biological and chemical contamination also need proper detection for various food applications [35].

Chemical and bionanosensors are the devices used to sense chemical and biological contaminants, respectively, as shown in Fig. 21.3. Toxicants and pathogens are biological contaminants, while chemical contaminants include pesticides, veterinary drugs, and food additives. Natural toxicants, such as aflatoxins and ochratoxins, and foodborne pathogenic bacteria, such as *Salmonella*, *Campylobacter*, *E. coli*, *Listeria monocytogenes*, *Bacillus cereus*, *S. aureus*, and other bacteria which are produced during making, distribution, and conservation of food are the major sources of biological contaminants [36]. ELISA is one of the available methods to detect bacterial toxins in a food product. It is limited to sensing only and cannot be used to observe the functional activity of toxicants. Another method to detect the toxigenic effect of bacteria is based on DNA probes. This method has been found highly accurate and sensitive, but its positive result cannot confirm that toxic genes have been expressed in strains of bacteria. To overcome these limitations the nanomaterials are fabricated in various sensors to enhance food safety and quality. Development of economic and efficient nanosensors is in demand, which can replace conventional methods such as chromatography and spectroscopy techniques used for quality analysis in the food industry. Since very few nanosensors have been found useful for process and quality control of food, continuous efforts are required to fabricate low-cost and highly sensitive nanosensors

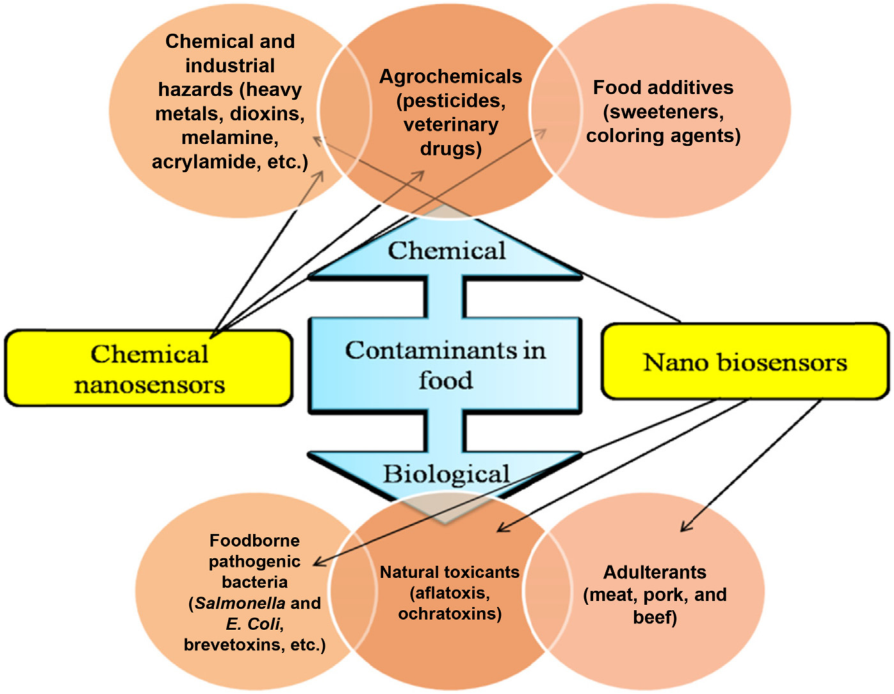

FIGURE 21.3 Nanosensors to detect various contaminants (nanobiosensors show better sensing efficiency compared to chemical nanosensors). *Reprinted with permission from B. Kuswandi, D. Futra, L. Heng, Nanosensors for the detection of food contaminants, in: A. Grumezescu, A. Oprea (Eds.), Nanotechnology Applications in Food, Elsevier, 2017, pp. 307−333, 9780128119433. © 2017 Elsevier.*

and nanobiosensors that can be applied in real conditions [35]. There exist undesirable chemical substances as by-products of synthesis and distribution of food products, from polluted surroundings and other natural sources. Contaminants including chemicals used in agriculture, pesticides, heavy metals, and melamine exert a harmful effect on humans and pets. Common techniques used for sensing of chemical contaminants are high-performance liquid chromatography (HPLC), liquid chromatography/mass spectroscopy (MS), ELISA, capillary electrophoresis, and gas chromatography(GC)/MS. These methods require costly instruments and skilled personnel. In spite of having considerable accuracy and sensitivity, there is a limited practical application of these methods. In this case the nanosensors and nanobiosensors can be the easy alternative to provide accuracy and sensitivity [35].

21.2.1.1 Sensing preservatives

Preservation of food is one of the focused research problems in the food industry. In the past few years several technologies have been adopted for the preservation of food based on nanosensors. These include silicon-based microfluid system, the addition of chemicals, such as sorbate, benzoate, *p*-hydroxybenzoic acid esters, and glutamate to enhance the performance of the different types of electrophoresis methods in chip setup and devices made up by NEMS. These devices have received considerable commercial interest due to their nano- to millimeter-sized components used in food preservation. The Polychromix Inc. has developed DTS to detect *trans*-fat content in food using NEMS technology [37].

21.2.1.2 Sensing adulterants

Adulterants in food sometimes can give rise to chemical hazards, and thereby it is necessary to detect them. Melamine (1,3,5-triazine-2,4,6-triamine), an organic compound, is an example which is commercially synthesized from cyanic acid that releases from urea during the chemical reaction. It can cause serious longtime toxicity and urinary stone in animals [35]. Numerous pet deaths have been found due to adulteration caused by the presence of melamine in food. Generally, the measure of protein content in food is actually the detection of nitrogen portion. Hence, the amount of protein content in milk products can be increased by the addition of melamine that has 66% of the nitrogen in it. However, at the same time, excessive use of it can produce insoluble melamine cyanurate crystal. This adulteration causes severe damage to the kidneys as well as reproductive organs of animals [38].

The literature study shows that analytical techniques based on modern instruments and nanobiosensors are used to sense melamine in food [39]. Sensors fabricated using cadmium telluride (CdTe), quantum dots (QDs), gold NPs (AuNPs), and single-walled CNTs (SWCNTs) have been observed as prominent candidates for the fast sensing of melamine in food items [40]. A calorimetric method has been applied in a way that the presence of melamine is detected with a color modification of colloidal AuNPs solution from red to blue. It is an easy, efficient, and low-cost technique because no external devices are required and melamine can be detected with naked eyes [40]. A similar approach has been applied for rapid detection of melamine adulteration in milk and milk products, such as infant formula, milk powder, lactose, wheat bran, and whey protein [35].

Fluorescence-based techniques using NPs of CdTe QDs can also be offered for sensing of melamine [41]. Another technique implies the use of SWCNT and modified graphene with a glassy carbon. Though it is a sensitive method for melamine detection, due to economic concern, graphene has been partially replaced by AuNPs. It implies the scope of graphene applications in future [35]. Adulteration in certain foods, such as pork, processed beef, meat, and beef jerky, also needs attention in the global market [42]. Again calorimetric technique based on AuNPs is able to detect adulteration in pork, chicken meat, and beef by the change in color from pinkish-red to gray-purple [43].

21.2.1.3 Sensing heavy metals

Contaminants from surroundings, such as soil and environment, contain heavy metals. Serious health problems arise even if a small quantity of heavy metals is exposed to humans. Removal of waste products and water from industries is one of the reasons responsible for soil contamination. It has been observed that pollution due to chromium in some areas causes the accumulation of chromium ions within plants in that particular area. These ions are responsible for toxic effects on humans and living organisms when their threshold level exceeds. They can also create damage in the nervous system, liver, skin, bone, and teeth [44]. Frequent use of pesticides and fertilizers in agriculture also cause heavy metal contamination in the environment [45]. Recently, spectroscopic, voltammetric, and chromatographic techniques are used to detect heavy metals, since they are efficient to trace even small quantity in samples. These methods are costly; hence, practically it is difficult to implement at large scale [35]. Nanomaterials, when incorporated with biosensors, can improve the sensitivity of detection. These include multiwall CNTs [46], AuNPs [47], silver NPs (AgNPs) [48], silica mesoporous/nanospheres [49], and rGO [50].

Past study shows that nanomaterials release heavy metals responsible for toxic effects when settled in a food product for a long time [51]. Metal nanooxides of ZnO, Ag, and CuO have been found to increase the intracellular reactive oxygen species (ROS) level, which has an adverse effect on DNA and lipid content of cell membrane [52]. Presence of Cu, Mn, Pb, Co, Cd, and Ni in various samples from environment can be detected by silica-modified magnetic NPs (MNPs) that are coated with cationic surfactant [53]. Among all other nanomaterials, MNPs have received attention due to their economic performance for the separation of heavy metals [54]. A recent study shows that magnetic iron oxide NPs such as Cu^{2+}, Ni^{2+}, Pb^{2+}, and Zn^{2+}, when aminated, can behave as an adsorbent to extract aqueous heavy metal ions from the sample [55]. MgO NPs prepared by sol−gel and calcination method have found capable to clear heavy metals from contaminated water sample as well as to minimize the effect of bacterial infection [56]. Synthesized iron oxides those are reusable and recoverable showed their stability when reducing heavy metals from the sample [57].

21.2.1.4 Detection of veterinary drugs

Nowadays, the presence and sensing of veterinary drug residues in food are important health concerns, particularly when consumed in excessive quantity. These drugs cause the production of bacteria in the body that can resist the effect of antibiotics or biocides [35]. It is reported that drug sensitization causes an allergic reaction and consequently a disturbance in the equilibrium of the natural ecological system [58]. To prevent the growth of bacteria, such as *Haemophilus influenzae*, *Neisseria gonorrhoeae*, *E. coli*, and *Shigella*, in medicine and agriculture, the penam group β-lactam antibiotic called ampicillin is used. Health problems, such as an allergic reaction, breathing difficulty, and seizures, in humans may occur because of the harmful residues in food products, which have emerged due to the improper use of ampicillin. Presence of ampicillin in food products has been identified by apta-nanosensor. Apta-nanosensor is a type of biosensor based on dual fluorescence calorimetric technique. It uses an aptamer based on AuNP solution to distinguish ampicillin among other antibiotics. When a specific single-stranded DNA aptamer called AMP17 comes into contact with ampicillin, it liberates AuNPs. Aggregation of these AuNPs reacts with salt and changes the red-colored solution to purple. This method is an accurate and highly sensitive alternative to detect ampicillin antibiotic in food products [35].

Kanamycin is another common antibiotic veterinary drug that is used in the treatment of septicemia, pneumonia, intestinal infections, and urinary tract infections [59]. Song et al. reported the fabrication of AuNP-based aptasensor to detect kanamycin as shown in Fig. 21.4. Again it is a calorimetric technique−based nanosensor in which the

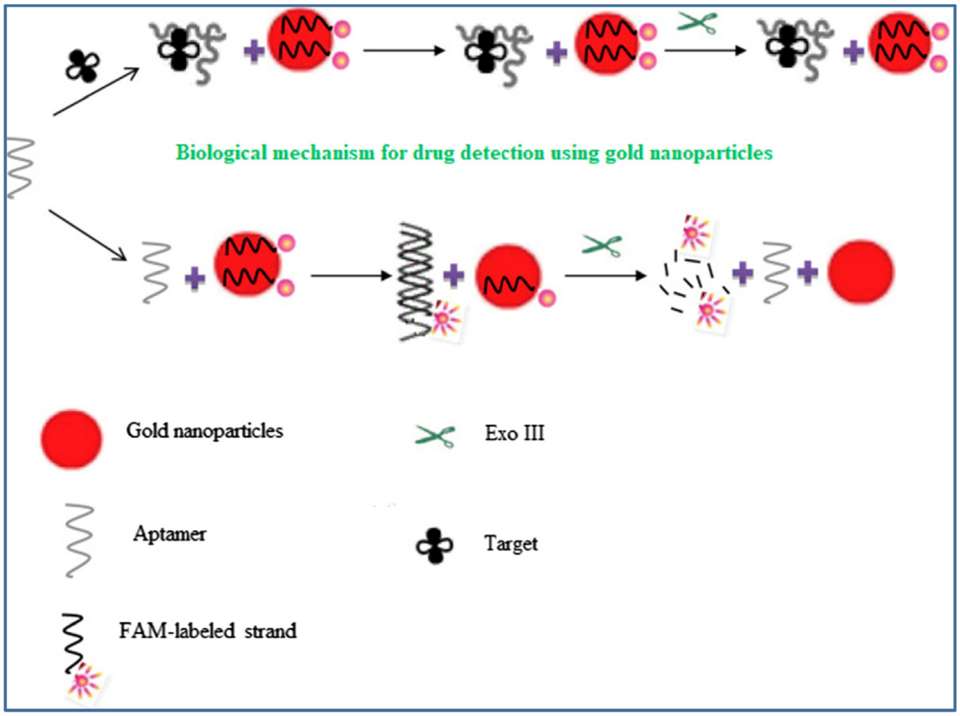

FIGURE 21.4 Mechanism of fluorescence assay technique to detect kanamycin using gold nanoparticles (AuNPs). *Reprinted with permission from M. Ramezani, et al., A selective and sensitive fluorescent aptasensor for detection of kanamycin based on catalytic recycling activity of exonuclease III and gold nanoparticles, Sens. Actuators, B: Chem. 222 (2016) 1–7. © 2016 Elsevier.*

aggregation of NPs when reacted with salt causes change in color in the presence of kanamycin [60]. It is found in the literature that the detection of kanamycin using calorimetric-based techniques is simple and economic since the change in color can be observed directly from naked eyes [60–65]. Furthermore, aptasensors can also be fabricated using the fluorometric techniques. Applications of different NPs such as silica NPs, amino-Fe_3O_4 MNPs, AuNPs, and upconverting NPs in this process enhance sensitivity due to their unique properties at nanoscale. Amplification of fluorometric signals in the detection of kanamycin has been achieved by CNTs and reduced rGO. Fluorometric aptasensors developed using molybdenum disulfide (MoS_2) nanosheets and carbon dots exhibit better quenching ability compared to graphene and AuNPs [66,67]. Tetracycline antibiotics have various applications in veterinary and aquaculture medicine. The residues of these antibiotics have been found in soil, water, and other environments. Detection of tetracycline can be done through liquid chromatography method based on MNPs. Efficiency to detect a quantity of tetracycline using this method is 86.2–4000 μg/kg [68].

21.2.1.5 Detection of growth-promoting agents and steroids

Steroids are natural hormones and contain lipid compounds. In adrenal cortex, more than 30 steroids are produced. Steroids can be classified basically into three categories based on their functional properties. Glucocorticoids, androgens, aldosterone, 11-deoxycortisol, cortisol, mineralocorticoids, corticosterone, and 11-deoxycorticosterone steroids are made in adrenal gland, whereas other steroid hormones such as estrogen are synthesized in the adrenal glands and gonads [69].

The level of steroid concentration is crucial in the body since the higher concentration of this hormone causes endocrinological disorder related to adrenal and gonadal function. The concentration of steroid hormones such as estradiol is very low in adult, which is up to 110 pmol/L, whereas, for children, this concentration is 18–165 pmol/L. In the case of progesterone hormone found in women, it is 3.2 nmol/L, while testosterone concentration is 0.35–5 nmol/L in men. Traditionally, immunoassays such as radioimmunoassay are used to detect hormones. Owing to the limitations of GC, to determine and lower the concentration of natural steroids, mass spectrometry has been used recently. Further, due to the typical procedure, this method was replaced with liquid chromatography–mass spectrometry method [70]. Recently, a variety of electrochemical sensors and its modifications based on pyrolytic graphite, carbon paste, molecularly imprinted polymers, CNTs, C60, and glassy carbon nanocomposites have been fabricated to detect steroids. Three steroids, namely, hydrocortisone, dexamethasone, and prednisolone, can be detected using differential plus voltammetry, when carbon paste and modified carbon paste electrode are present. SWCNTs, modified with edge plane pyrolytic graphite electrode, are utilized to detect betamethasone steroids. The effective surface area leads to enhanced detection limit up to 25 times of modified nanosensor in comparison to bare nanosensors [71].

The new class of biosensors was formulated to increase the accuracy in the detection of lower concentrations of steroids in the body called nanobiosensors. Boron-doped silicon nanowires and CNTs have been in use as chemical and biological sensors. For steroids, detection isomerases were used as receptor due to well-understood enzyme function. The primary concept of sensing mechanism is the well-structured intermolecular binding of a charged ligand to replicate the overall binding of the analyte to the protein residues. The hydrophobic effect creates large driving forces to promote proper binding to the hydrophobic ligand with protein. By controlling the enthalpy, thermodynamics of protein−ligand binding can be changed very well. Analyte replaces the situated ligand and gets exposed on the surface of the SiNW. In the presence of asteroid the negatively charged 5-(2-aminoethyl amino)-1-naphthalenesulfonic acid (1,5-EDANS) moiety presumably resides at the steroid-binding site and is expelled to get exposed to the nanowire surface. The electrical output generated from the 1,5-EDANS moiety is measured to determine concentration with the sensitivity in the range of femtomolar levels [72].

21.2.2 Detection of food spoilage and moisture

Spoilage of food has received considered attention in food safety as it has direct impact on public health. Nanosensors are fabricated in such a way that constituent NPs fluoresce in different colors when exposed to food pathogens. When spoilage occurs in the food product, it starts to release certain chemicals. "Electronic tongue" and "electronic nose" are the specially designed nanosensors which detect these chemicals when used in food packaging [73]. In addition to this, food containers coated with AgNPs have been put forward to prevent spoilage in food. Remarkable application of nanosensors is the ability to measure food spoilage in the packet of processed food [74]. Exposure to certain gases can also result in spoilage of food products. It can be detected using nanosensors and prevented by proper treatment and packaging made up of biodegradable material to avoid environmental pollution [75−77]. For this, gas nanosensors are used, which can generally trace the presence of hydrogen sulfide, hydrogen, nitrogen oxides, ammonia gases, and sulfur dioxide [78−80]. Gas sensors fabricated using palladium, platinum, and AuNPs are sensitive and efficient compared to other conventional nanosensors [81−83]. Carcinogenic toxins, such as aflatoxin B1 present in milk, can be sensed using nanosensors based on AuNPs [75].

21.2.3 Detection of sweeteners

A key application of nanosensor in food processing is the development of nanostructured products, such as spreads, mayonnaise, yogurts, and ice creams. In nanoscale processing the nanostructured food products are demanded to possess new tastes, textures, consistency, and emulsion stability, related to conventional approaches. Nanostructured food is a low-fat product as compared to cream-based products and offers the consumers a "healthy" option without compromising on the taste. Nanostructured mayonnaise is synthesized through a collection of nanomicelles comprising nanosized water drops. The mayonnaise offers the same flavor and appearance qualities as its full-fat equivalent, but with a considerably less amount of fat. Another application of nanotechnology is in food additives and supplements by using the concept of nanoencapsulation. This concerns with the formulation of additives into liposomes or encapsulates with phospholipids, proteins, or other degradable/digestible polymers. Nanosensors are likely tools for the application in food production and agriculture. In comparison to traditional chemical and biological methods for selectivity, nanosensors offer high response time and sensitivity. To replace sucrose, different types of compounds are used as sweeteners. Sweeteners can be of two types: intense and bulk sweeteners. Intense sweeteners are obtained by chemical methods and thus called artificial sweeteners. They are largely used in powdered drink, carbonated beverage, candy, and desserts. The commonly used examples of such sweeteners are aspartame, sucralose, neohesperidin, thaumatin, cyclamate, saccharin, and acesulfame. It was found out through in-depth studies that these intense sweeteners create health problems such as obesity, dental issues, and cancer, as well as are also responsible to increase blood sugar level. On the other hand, bulk sweeteners have reduced the amount of sweetness compared to intense sweeteners, and these are generally used in several food products such as sweets, jam, preservatives, baked items, ice cream, and cereals except soft drinks. Bulk sweeteners, such as mannito, maltitol, isomalt, lactitol, erythritol, and sorbitol, are basically hydrogenated carbohydrate with additional functions in comparison to sucrose such as lowering the freezing point and reduce the caramelization of ice cream. The energy contribution of bulk and intense sweeteners is 10 kJ/g, whereas the contribution of sucrose in food energy is 17 kJ/g [84]. Traditionally, HPLC with UV was used to detect these sweeteners, but all these methods are complicated [85]. However, the lipid membrane−based nanosensors were effective enough to detect artificial sweeteners, such as acesulfame K, cyclamate, and saccharin. Saccharin is prohibited in many countries owning to its carcinogenic property.

The catalytic hydrogenation of neohesperidin yields neohesperidin dihydrochalcone (NHDC) that is also a commonly used artificial sweetener. It is 20 times sweeter than saccharin, and 1500−1800 times sweeter from sucrose. It is widely used for the pharmaceutical industry since it cuts down the sensitivity to the bitter taste perceived by human taste sensory buds. In accordance with the scientific committee for foods of the European Community in 1987, the maximum permissible limit of any food is bound by certain regulations for example, an additive is 5mg/kg. Of all the methods present to detect this compound such as HPLC, electrochemical and spectroscopic methods draw maximum attention of researchers due to its fast response, time saving, low-cost instrumentation, easy to operate, and high sensitivity and selectivity. However, due to the high electrical conductivity and large surface area, CNTs have been incorporated within the electrochemical analysis. SWCNTs modified with a glassy carbon electrode was also found to be effective as voltammetric sensor demonstrating high sensitivity response for NHDC. Exceptional results for this were exhibited in the presence of phosphate buffer solutions [35].

21.2.4 Detection of genetically modified food

Australian Federal Government Agency prohibited the whole cultivation of genetically modified wheat with altered gene structure after conducting various research on its disadvantages. The Philippine government's International Rice Research Institute conducted research on golden rice in August 2013, where it was found that the product demonstrated an increased amount of certain protein residues that are the precursors of vitamin A [86]. A biological technique called genetic modification has been employed to alter the genetic makeup of living organisms. According to WHO (World Health Organization), this can be achieved by modifying genetic material (DNA) that does not occur naturally by mating. Genes can be transferred from one organism into the other that is unrelated to each other. According to European Commission and FAO (Food and Agriculture Organization of the United Nations), genetically modified organisms does not occur naturally [87]. Similarly, genetically modified food is also produced through genetically modified plants and animals. Triticale is widely used in pasta and grains that is produced by crossing of wheat and rye. The resulting hybrid is sterile.

Genetically modified foods are produced by introducing the gene coding for certain trails into a plant cell, after that, they repeat generation of plants by tissue culture. There are certain techniques available to modify genes into the cell. Among all these techniques, microparticle bombardment is used widely. In this technique the naked DNA is substrated on tungsten and gold microparticles and then targeted on tissues such as embryonic tissue from seed or meristems with high velocity (propelled by pressurized helium). Electroporation technique is used to modify genes into the cell that involves the motion of negatively charged DNA into the electric potential gradient. Application of mesoporous silica NPs and silica carbide slivers in chloroplast transformation by microinjection is another example of such techniques [88]. Among all these techniques, microparticle bombardment of DNA is the most efficient technique. Genetically modified food creates an adverse effect on the human body. Genetic hazards, toxicity, and allergy are the major risks associated with genetically modified foods [87].

21.3 Smart food packaging sensors

Nanosensors are tiny, precise, and portable devices, developed for the sensing of physical quantity of materials and converts them into measurable signals. Their main application in food industries is to sense quality and other parameters such as moisture, adulterants, amount of preservatives, and durability of the products. Previous study reveals that the current smart packaging segment is influenced by oxygen scavengers and moisture absorbers accounting for major products of the market.

Nanoenabled packaging technology has been widely applied to bakery and meat products as shown in Fig. 21.5. Food preservation is the food treatment to avoid alteration in its original properties. Other food preservation conventional methods are canning, freezing, and drying. It is reported that the nanotechnology suggests effective and more authentic way to improve food preservation by the use of nanosensors, nanocomposites, and NPs in food packing [89,90]. Nanosensors support in the revealing of any delicate changes in food color and the presence of gases generated because of food spoilage [91−93]. Moreover, nanosensors are highly sensitive and selective to these modifications making them more capable than the traditional approaches of sensors [94−96]. The gas sensors are made up of transition metals such as palladium, platinum, and gold. The presence of aflatoxin B1 toxin has been observed in milk, and it can be detected by gold metal−based NPs [97]. The packaging based on SWCNTs and DNA is capable of a significant

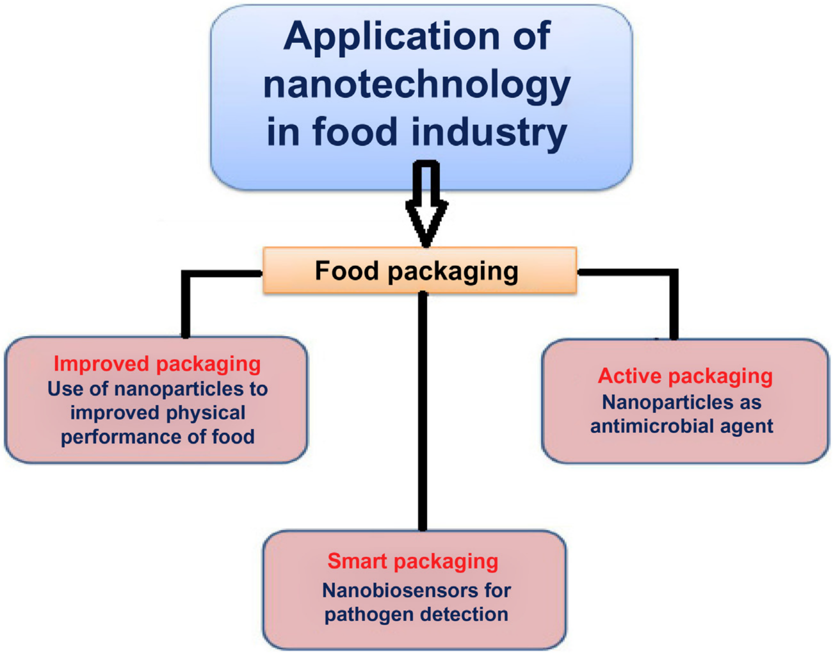

FIGURE 21.5 Role of nanotechnology in food packaging. *Reprinted with permission from T. Singh, et al., Application of nanotechnology in food science: perception and overview, Front. Microbiol. 8 (2017) 1501. Licensed under Creative Commons Attribution License (CC BY).*

increase in the sensitivity of sensors. In agriculture, nanosensors are used to detect the presence of pesticides on the surface of fruits and vegetables. Certain nanosensors have also been reported to detect food products containing carcinogens [98].

In nanocomposite-based food packaging a polymer can serve as a better alternative of generally used packaging materials such as paper, glass, and metals because of their low-cost and efficiency. Nanocomposites are a polymer matrix reinforced in the nanofillers such as cellulose microfibrils, CNTs, nanoclays, and nanooxides. Several examples of such widely used synthetic and natural polymers in the packaging of food are polyamide, polystyrene, nylon, polyolefins, chitosan, cellulose, and carrageenan. Nanocomposites can be employed in preserving the freshness of food items for comparatively longer time period irrespective of the infestation by microbials present in the food products. They reduce carbon dioxide (CO_2), escaping from carbonated drinks containers by performing as gas barriers [99]. Since the food environment is continuously detected for temperature and pathogens, effective oxygen content indicators are widely used. They indicate lifetime of the products using nanosensors. AuNP-based enzymes to detect microbes, gas sensing to detect food condition, and nanofibrils of perylene-based fluorophores to sense spoilage in meat and fish by detecting gaseous amines are examples of the applications of nanosensor in smart food packaging [24,100]. Titanium oxide and zinc oxide nanocomposites have also been studied to detect volatile organic compounds [80]. Advantages of metal oxides (TiO_2 and ZnO NPs) in food preservation are chemical and physical stability, low-cost, easy availability, and nontoxic for humans [101–103]. TiO_2 NPs used in different packaging materials exhibit significant photo degradability and antibacterial property. In addition, these composite materials were revealed to control Gram positive and Gram negative microorganisms, such as *S. aureus, E. coli, Bacillus subtilis, Klebsiella pneumoniae, Saccharomyces cerevisiae, and Pseudomonas Aeruginosa*, without the need of being released to the medium [104]. The antibacterial performance of TiO_2 on microorganism is due to the damage in membrane and cell wall [105].

Important advantages of NPs in food processing are improved food texture and quality. Earlier, NPs have been applied in drug delivery; now in a similar manner, they are used in the food sector, and their effects depend on their bioavailability. Silicate and other NPs are reported to control the flow of oxygen in packaging materials. Moisture leakage can also be detected and controlled so that food maintains freshness for a comparatively longer time. During the process, certain nanoparticles that are selectively attached to pathogens are effectively removed [106]. The recent concept of nanobarcodes used for tagging improves security systems [107]. While the application of smart sensors is advantageous to the consumers in view of better quality, it also provides an effective alternate to consumers for fast distribution and validation of the food products [108].

21.4 Nanosensors for product acceptability

Acceptance of products by consumers and industries depends on various factors such as the safety of food, proper packaging, and stability from external surroundings including light, heat, humidity and oxygen level, enzymes, odors, relative pressure, microorganisms, contaminants, insects, gaseous emissions, and lipid oxidation. Packaging of food products plays a crucial role in conservation, quality maintenance, and stability, which should be as per consumer demand and acceptable from human health perspective [109]. Consumer acceptance plays a vital role to determine commercial success as well as social advantages of any recently implemented technology [110,111]. Food Safety Authority of Ireland (FSAI) has reported that nanotechnology has significant potential in food industries. It can be implemented for the emulsion and formulation of encapsulation in food contact materials and in the development of nanosensors [112]. Various benefits of the use of nanotechnology in food sector have been observed, which can have a great impact on its commercial acceptability. Several industries have already accepted nano-based technologies in different food-related applications such as enhancement of flavor and color, change in its texture, maximum absorption of nutrient substance, smart packaging of food products to prolong its lifetime, and detection of harmful bacteria using nanomaterials-based bio/chemical sensors. Kraft food is a United States based food manufacturing and processing company that established Nanotechnology Consortium in 2000 to produce food products as per individual consumers' requirements. They include different factors such as allergies and reaction of a certain food, and nutritional need. Research is still ongoing to improve food textures by Nestle and Unilever food industries. Omega-3-fatty acid is an essentially required nutrient for the human body. It is provided using nanocapsules by the well-known food industry in Australia. Development of non-toxic nanoscale herbicides to attach weed's seed coating and preventing seeds from germinating has been studied in Asia by collaborative efforts of India and Mexico [113]. In past study, Weiss et al. mentioned advantages of the use of nanotechnology in food industries. In this regard, it is observed that the produced food can have less amount of fat and sugar as well as modified taste and textures [114]. In addition, nutrient consumption, packaging, and food safety can also be improved [115].

21.5 Conclusion

Factors, such as safety, hygiene, proper packaging, providing conducive external environmental parameters, optimum exposures to light and temperatures, oxidative stress, flavor regulations, contaminants, gaseous emissions, and lipid oxidation, mainly govern the success and acceptability of any food industry in the market, directly influencing the brand value. For these reasons, various packaging techniques applied play an instrumental role in the conservation of the quality and the required considerations for the acceptability of food products by the consumers. The enhanced properties of these molecular wonders in the scale of 1−100 nm have proved to be instrumental in upkeeping the required industrial procedures regarding production, process, and packaging. In the current age the successful use of smart nanoenabled packaging techniques loaded with oxygen and moisture scavenging properties makes the role of nanotechnology vital in the field of food industry. The high sensitivity of nanomaterials as effective sensors of various chemical and biological contaminants in the food sector is making these dynamic synthetic molecular frameworks as an irreplaceable representative in the emerging field of food sector.

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